

Connor Sequeira

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Mechatronics Engineering student focused on robotics, automation, embedded systems, and AI/vision-enabled inspection workflows.

EDUCATION

BASc Mechatronics Engineering | University of Waterloo - Waterloo, ON
Minor in Artificial Intelligence

Expected 2028

TECHNICAL SKILLS

Programming & Tools: C++, Python, JavaScript, HTML/CSS, Git/GitHub

AI & Vision Software: PyTorch, OpenCV, model training/inference, computer vision, AI inspection workflows

Embedded & Automation: Arduino, embedded C++, ultrasonic sensing, LCD/servo integration, I/O control, robot logic/flow

Robotics & Design: UR5e, FANUC CRX, Mech-Mind 3D vision, SolidWorks, AutoCAD, 3D printing, SimScale/FEA

PROFESSIONAL EXPERIENCE

Robotics & Automation Co-op / Engineering Analyst | A. Berger Precision - Brampton, ON
3 co-op terms, 12 months total: Sep-Dec 2024, May-Aug 2025, Jan-May 2026

Sep 2024 - May 2026

- Developed/debugged UR5e pick-and-place logic, reducing cycle time from 21.3s to 6.7s through motion tuning and sequence optimization.
- Supported programming, training, configuration, and validation of an AI vision inspection system, including image capture setup, model-training iterations, threshold tuning, and pass/fail testing.
- Constructed Mech-Mind 3D camera workflows and robot-vision integration logic to improve pick reliability and reduce part mispicks during operation.
- Assisted FANUC CRX-5iA robot cell setup for AI-powered inspection, including robot motion, safety-device integration, and inspection workflow testing.
- Designed SolidWorks parts, tooling, and 3D-printed prototypes; used simulation/FEA where useful to improve durability and support production fixes.
- Created documentation, flowcharts, and troubleshooting guides to improve robot-cell maintainability and reduce operator knowledge gaps.

Engineering Assistant | Armacell Canada, Brampton, ON

Jan 2024 - Apr 2024

- Designed a SolidWorks printer mount for product-traceability needs while accounting for operating space, safety, and installation constraints.
- Prepared cost-benefit analysis, risk assessment, and technical report for a proposed solution to leaking blender machines.

SELECTED PROJECTS

Smart Garage Door Access System | 2026 | [Arduino](#), [C++](#), [ultrasonic sensor](#), [LCD](#), [servo control](#)

- Built an Arduino-based garage door prototype that detects an approaching vehicle with ultrasonic sensing, opens a servo-controlled door after a stable 2-second detection, and displays door status on an LCD.
- Implemented state-based control logic for CLOSED, OPENING, OPEN, and CLOSING states with red/yellow/green indicators and buzzer warning during door closure.

PyTorch Digit Recognition Classifier | 2026 | [Python](#), [PyTorch](#), [machine learning](#), [model training](#)

- Trained and evaluated a neural-network classifier for handwritten digit recognition, including dataset loading, training loop, validation metrics, and inference testing.
- Saved the trained model and created an inference script/UI workflow to test predictions on unseen handwritten digit inputs.

TRON Haptics | 2025 | [C](#), [Python](#), [IMU integration](#), [motor control](#), [3D printing](#)

- Developed a wearable haptic glove that dynamically adjusts resistance to simulate touch sensations in virtual environments.
- Integrated embedded control, motor-driven resistance, sensor input, and 3D-printed mechanisms into a functional wearable prototype.

PrecisionMOTION Robotic Hand | 2024 | [C++](#), [Python](#), [OpenCV](#), [embedded systems](#), [SolidWorks](#)

- Created a functioning replica hand that mimics user inputs by combining mechanical design, sensor feedback, embedded control, and software integration.
- Designed and prototyped mechanical components while integrating electronics and control logic for real-time hand motion response.

AUTOlathe | 2023 | [RobotC](#), [SolidWorks](#), [sensors](#), [servos](#), [mechanical design](#)

- Designed a model lathe that follows user-drawn profiles using mechanical design, sensor feedback, and servo-based actuation.
- Built and tested the prototype logic and mechanical layout to demonstrate automated profile-following on a small-scale lathe concept.